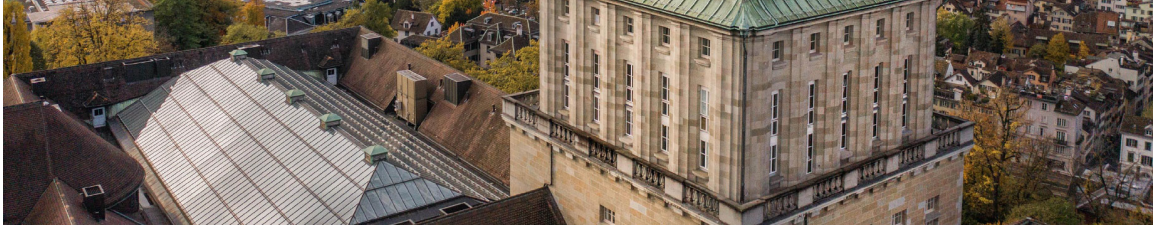


Towards Bayesian joint modelling of ranks and scores to reduce uncertainty in the assesment of grant proposal quality

Alvin Walker, supervision: Dr. Rachel Heyard & PD Dr. Malgorzata Roos

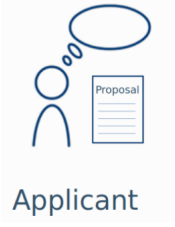
June 4, 2026



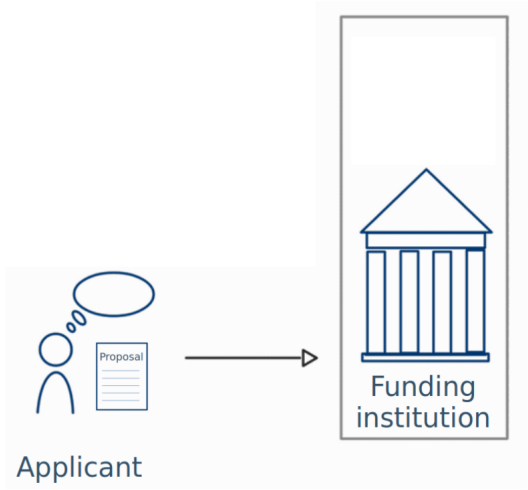
Presentation outline

1. **Context**
   - Grant funding allocation
   - Scoring and Ranking data
   - Thesis objectives
2. **Data exploration**
   - Scoring data
   - Ranking data
3. **Modelling**
   - Models for scoring data
   - Models for ranking data
   - Intermediate results
4. **Outlook**

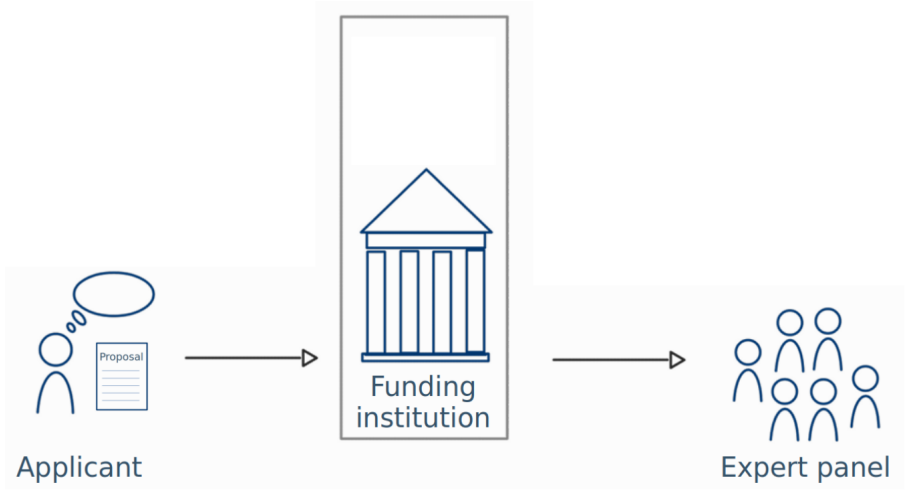
System for grant funding allocation



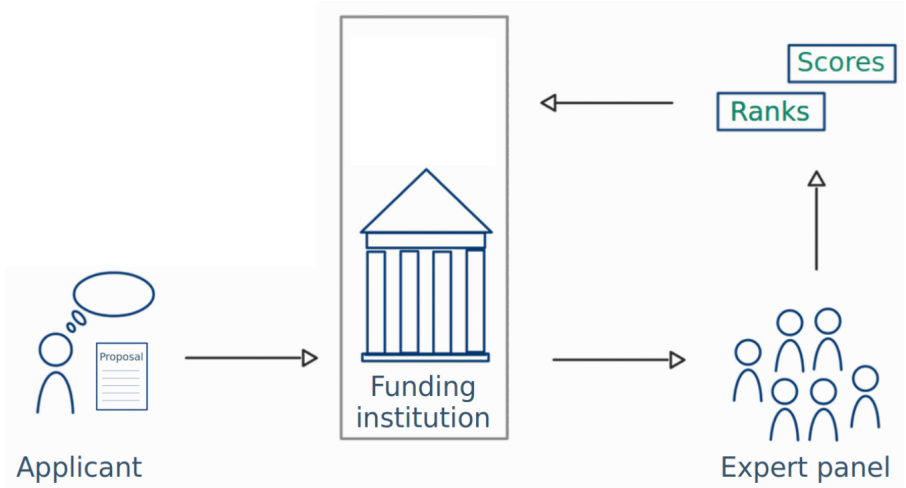
System for grant funding allocation



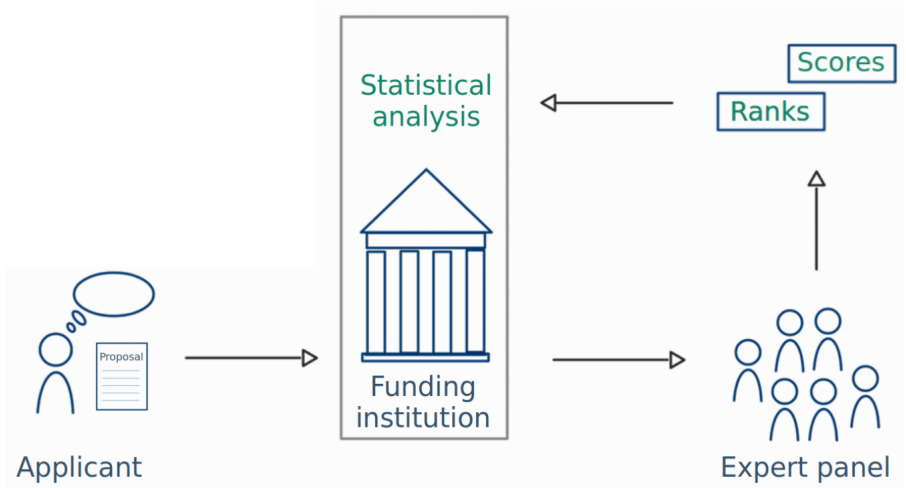
System for grant funding allocation



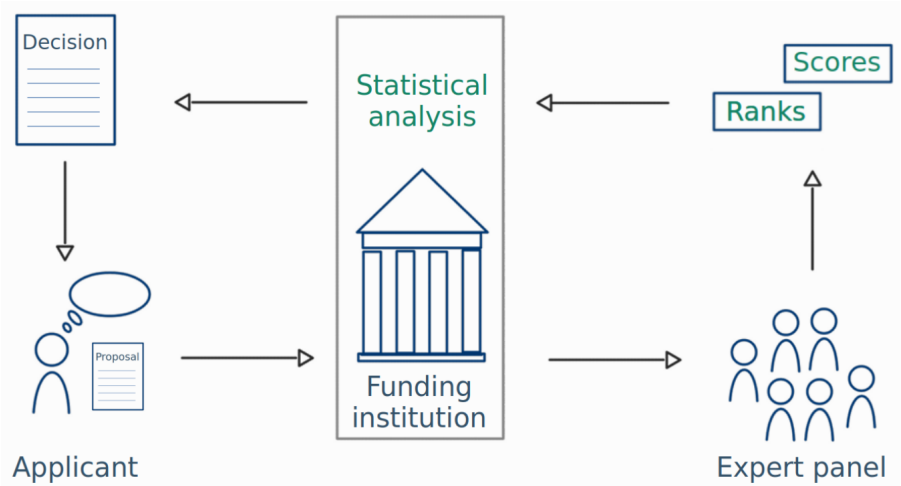
System for grant funding allocation



System for grant funding allocation



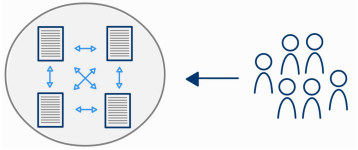
System for grant funding allocation



Panel review data

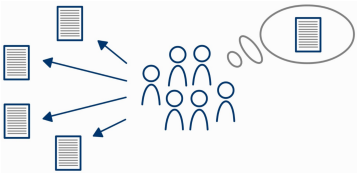
Ranking data

- Proposals are ranked on a relative scale.
- Either all or only top-k proposals are ranked.
- The assessment of each proposal depends directly on the other proposals.



Scoring data

- Proposals are scored on an absolute scale.
- This scale can either be ordinal or continuous.
- The assessment of each proposal should be independent from all other proposals.



Thesis objectives

"Towards Bayesian joint modelling of ranks and scores to reduce uncertainty in the assessment of grant proposal quality"

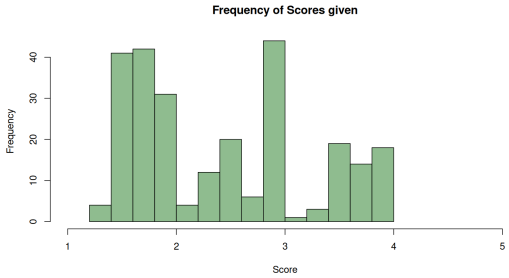
1. Identify and describe existing methods for quality assessment using scores and ranks.
2. Reproduce these methods on a data set from Gallo et al. (2023).
   - Scenario: Select the five best proposal from the data set.
3. Extend the Bayesian hierarchical model from Heyard et al. (2022) to jointly model ranks and scores and reduce the uncertainty in the assessment of grant proposal quality.

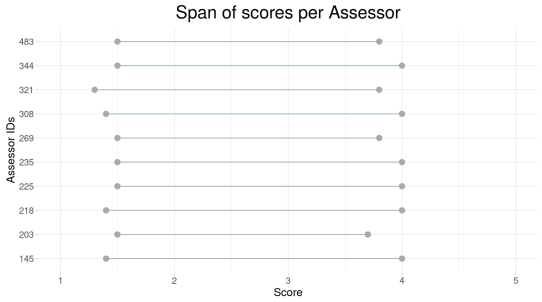
Overview of data (Gallo et al. 2023)

Property	Value
Scoring scale	1 to 5
Objects	28
Assessors	10
Scores per object	7-10
Total scores	259
Mean score	2.481
Standard deviation	0.82

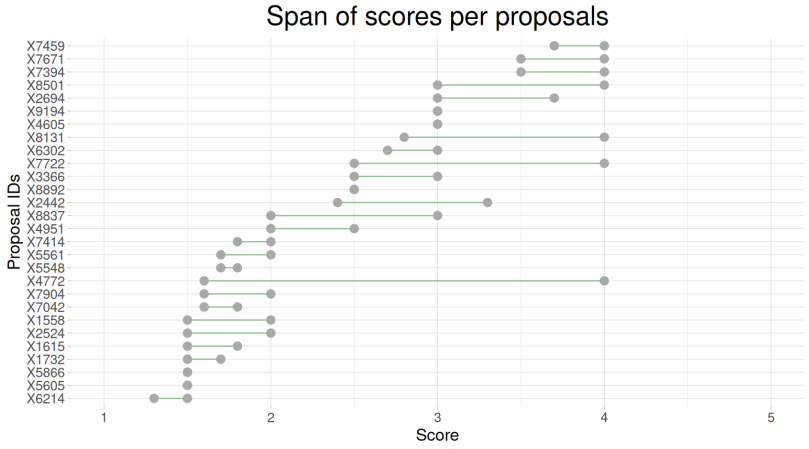
Property	Value
Ranking scale	top-6
Objects	11
Assessors	10
Ranks per object	1-10
Total ranks	60

Overview of scoring data (1/2) (Gallo et al. 2023)

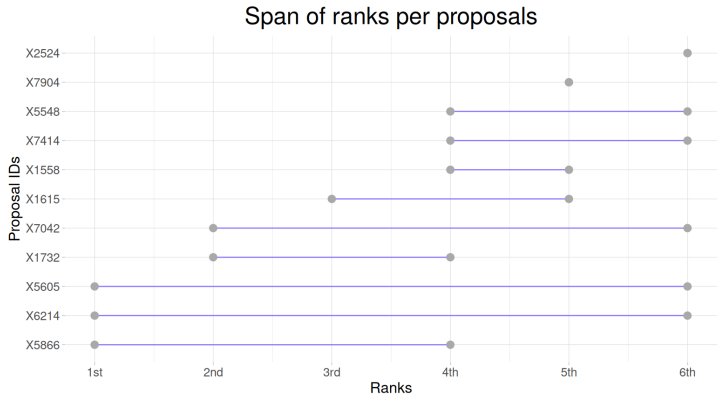




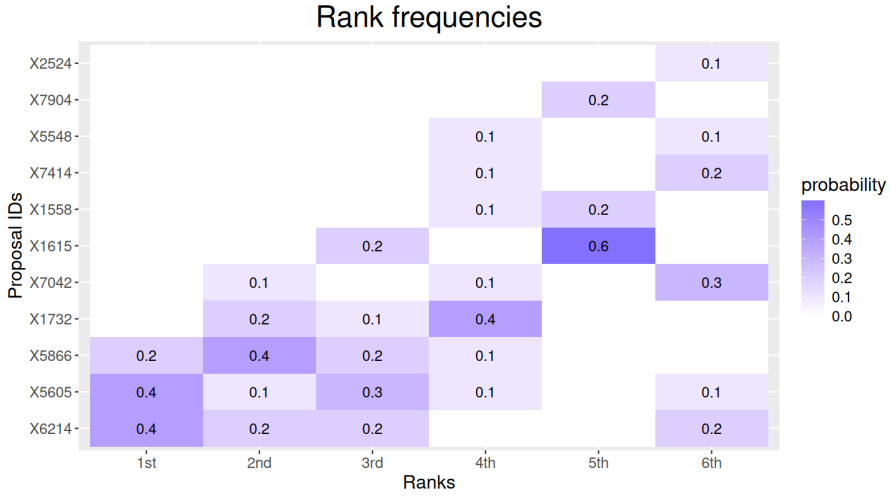
Overview of scoring data (2/2) (Gallo et al. 2023)



Overview of ranking data (1/2) (Gallo et al. 2023)



Overview of ranking data (2/2) (Gallo et al. 2023)



Model overview

	Scores	Ranks	Frequentist	Bayesian
Mean scores	✓		✓	
Hierarchical model for scores	✓			✓
Plackett-Luce		✓	✓	
Mallows		✓	✓	
Thurstonian		✓	✓	
Mallows-binomial	✓	✓	✓	
Joint hierarchical model	✓	✓		✓

Model overview

	Scores	Ranks	Frequentist	Bayesian
Mean scores	✓		✓	
Hierarchical model for scores	✓			✓
Plackett-Luce		✓	✓	
Mallows		✓	✓	
Thurstonian		✓	✓	
Mallows-binomial	✓	✓	✓	
Joint herarchical model	✓	✓		✓

Definitions - Modelling Scoring data

We define Alternatives, scores and quality, according to Xia et al. (2019) and Heyard et al. (2022)

$$\mathcal{A} = \{o_1, o_2, ..., o_n\}$$
$$y_{i,j} \rightarrow \text{score given to object } o_i \text{ by assessor } j$$
$$\theta_i \rightarrow \text{ quality of } o_i$$

Modelling approach

- We assume the scores $y_{i,j}$ are associated with a latent quality variable θ_i.
- We infer θ_i using multiple methods and sort it to create a ranking of objects.

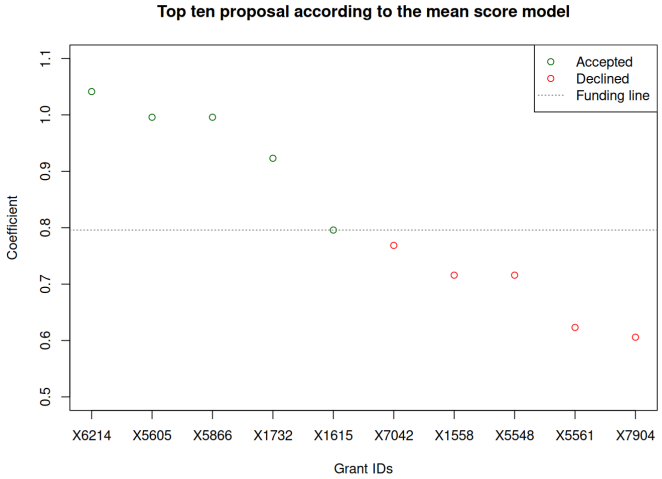
Definition: Mean score model

1. Model the quality θ_i through the mean score model as defined in Zehlike et al. (2022):

$$\bar{y}_i = \frac{\sum_{j=1}^{m} y_{ij}}{m}$$
$$\bar{y} = \frac{\sum_{i=1}^{n} \bar{y}_i}{n}$$
$$\hat{\theta}_i^{mean} = \bar{y}_i - \bar{y}$$

2. Infer θ_i^{mean} and sort in descending order to obtain final ranking.

Result: Mean score model



Shortcomings of Mean score model

1. No measures of uncertainty. (Kaplan et al., 2008)
2. Assumes assessors have the same scoring habits.
3. Assumes subset of proposal assigned doesn't effect scoring.

Definition: Bayesian Hierarchical Scoring

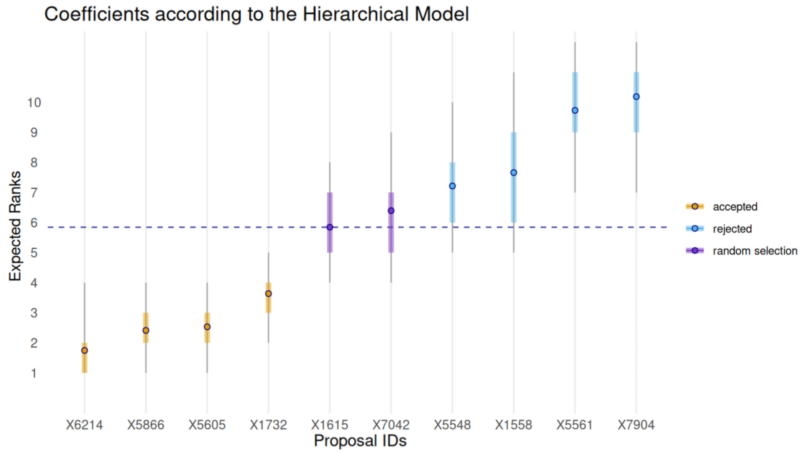
1. Model the quality θ_i through the Bayesian Hierarchical model as defined by Heyard et al. (2022):

$$
\begin{aligned}
y_{ij} | \theta_i^{hier}, \lambda_{ij} &\sim N(\bar{y} + \theta_i^{hier} + \lambda_j, \sigma^2) \\
\theta_i^{hier} &\sim N(0, \tau_\theta^2) \\
\lambda_j &\sim N(0, \tau_\lambda^2)
\end{aligned}
$$

2. Infer the posterior distribution of θ_i^{hier} using MCMC.

3. Use θ_i^{hier} to calculate the expected rank of each proposal.

Results: Bayesian Hierarchical Scoring



Modelling ranking data

We define ranking data according to to Xia et al. (2019) and Marden (2014):

$$\mathcal{L}(\mathcal{A}) \rightarrow \text{All possible orderings of the objects}$$
$$R_j \, \epsilon \, \mathcal{L}(\mathcal{A}) \rightarrow \text{Ranking order given by assessor } j$$
$$R(o_i) \rightarrow \text{Rank given to object } o_i$$

Example with three objects:

$$\mathcal{A} = \{o_1, o_2, o_3\}$$

$$\mathcal{L}(\mathcal{A}) = \begin{Bmatrix} [o_1 \succ o_2 \succ o_3] & [o_1 \succ o_3 \succ o_2] & [o_2 \succ o_1 \succ o_3] \\ [o_2 \succ o_3 \succ o_1] & [o_3 \succ o_1 \succ o_2] & [o_3 \succ o_2 \succ o_1] \end{Bmatrix}$$

$$R_1 = \{[o_1 \succ o_3 \succ o_2]\}$$

Definition: Plackett-Luce

Plackett-Luce models ranking data by estimating a set of weights $W = \{w_1, ..., w_n\}$ with a parameter w_i for each object o_i. (Plackett, 1975)

$$\mathbb{P}[R(o_1) = 1 \mid W = \{w_1, \dots, w_n\}] = \frac{w_1}{\sum_{i=1}^{n} w_i}$$

Example:

$$\mathcal{A} = \{o_1, o_2, o_3\}$$
$$W = \{w_1 = 0.3, w_2 = 0.2, w_3 = 0.5\}$$
$$\mathbb{P}[R(o_3) = 1 \mid W] = \frac{0.5}{0.3 + 0.2 + 0.5}$$

Plackett-Luce

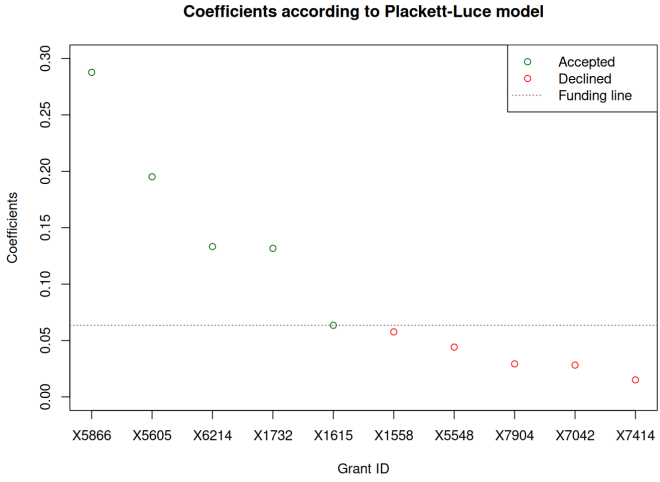
The Plackett-Luce model is sequential, it assumes the data is generated as follows,

1. Select an object from $\mathcal{A}$ to be in first position of the ranking using W.
2. Remove the weight from W and the object from $\mathcal{A}$
3. Select a new object from $\mathcal{A}$ to be in the next position of the ranking using W.
4. Repeat from step 2 until the full ranking is generated.

The probability of observing a ranking order can be written as follows,

$$\mathbb{P}[o_1 \succ \ldots \succ o_n \mid W] = \prod_{i=1}^{n} \frac{w_1}{\sum_{i=1}^{n} w_i} * \frac{w_2}{\sum_{i=2}^{n} w_i} * \cdots * \frac{w_n}{w_n}$$

Results: Plackett-Luce



Summary of results

Table of final rankings

	X6214	X5605	X5866	X1732	X1615	···
Mean score	1	2	3	4	5	···
Bayesian Hierarchical	1	3	2	4	5	···
Plackett-Luce	3	2	1	4	5	···

···	X7042	X1558	X5548	X5561	X7904	X7414
···	6	7	8	9	10	>10
···	5	8	7	9	10	>10
···	9	6	7	>10	8	10

Outlook

Modelling reproduction on Gallo et al.

- Mallows model for rankng data (Mallows, 1957)
- Thurstonian model for ranking data (Thurstone, 1927)
- Mallows-Binomial joint model (Pearce & Erosheva, 2022)
- Hierarchical joint modelling

Steps towards Hierarchical joint modelling

1. Translate the ERforResearch package from JAGS to INLA framework.
2. Implement a Bayesian imputation model for ranking data.
3. Implement a joint model for ranking and scores using the INLA joint modelling framework.

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Laird, N. M., & Louis, T. A. (1989).Empirical Bayes Ranking Methods. *Journal of Educational Statistics*, *14*(1), 29–46. https://doi.org/10.3102/10769986014001029 (siehe S. 33).

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References II

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Plackett, R. L. (1975).The Analysis of Permutations. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, *24*(2), 193–202. Verfügbar 4. Juni 2026 unter http://www.jstor.org/stable/2346567 (siehe S. 25).

Thurstone, L. L. (1927).A law of comparative judgment. *Psychological Review*, *34*(4), 273–286. https://doi.org/10.1037/h0070288 (siehe S. 29).

Xia, L., Brachman, R., & Rossi, F. (2019, Januar). *Learning and Decision-Making from Rank Data*. Morgan & Claypool Publishers.

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Questions and suggestions

Extra slide : Expected Ranks

We calculate the Expected ranks using the distributions of θ_i (Laird & Louis, 1989)

$$
\begin{aligned}
ER_i &= \mathbb{E}[R(\theta_i)] \\
&= \sum_{i=1}^{n} \mathbb{P}[\theta_i \leq \theta_j] \\
&= 1 + \sum_{j \neq i} \mathbb{P}[\theta_i \leq \theta_j]
\end{aligned}
$$

